

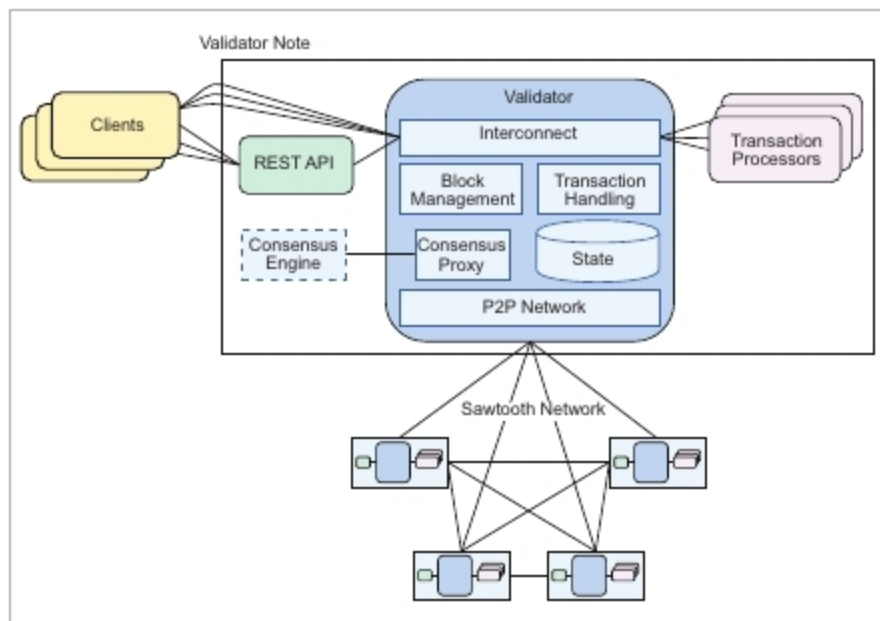


Figure 1: Hyperledger Sawtooth high-level architecture

Blocks and batches arrive via interconnect, either through the gossip protocol or from client requests. The processing of these blocks and batches is handled in multiple pipelines.

The completer initially receives the blocks and batches. It guarantees that all dependencies for the blocks and batches have been satisfied. The completer is responsible for making sure that blocks and batches are complete before delivering them to the BlockPublisher or ChainController. It also checks for dependencies, ensures that the previous block exists, and makes sure that the batches exist in the BlockStore or BlockCache. All blocks and batches have a timeout for being completed. The completer sends an initial request for any missing dependencies or predecessors. If a response is not received in the specified time, the block or batch is dropped. For example, consider the case of a chain A->B->C. If Block C arrives but B is not in the BlockCache, the completer will request Block B. If the request for B times out, Block C is dropped.

The BlockPublisher is responsible for creating candidate blocks to extend the current chain. It does all the housekeeping work for creating a block, but takes direction from the consensus algorithm for when to create a block and when to publish a block-based response to the following events:

- Start block
- Receive batch
- Summarise block (stop and make the block available)
- Finalise block (publish the block)

The ChainController is responsible for maintaining the blockchain for the validator. This responsibility involves validating proposed blocks, evaluating valid blocks to determine if they should be considered for the new chain head, and generating new blocks to extend the chain. It determines which chain the validator is currently on and coordinates any change-of-chain activities that need to happen.

The consensus interface is responsible for determining who can publish a block, whether a published block is valid according to the consensus rules, and which block should become the chain head in the case of a fork. Sawtooth

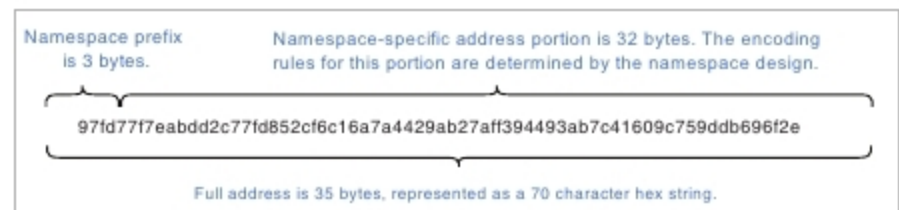


Figure 2: Radix addressing name spaces

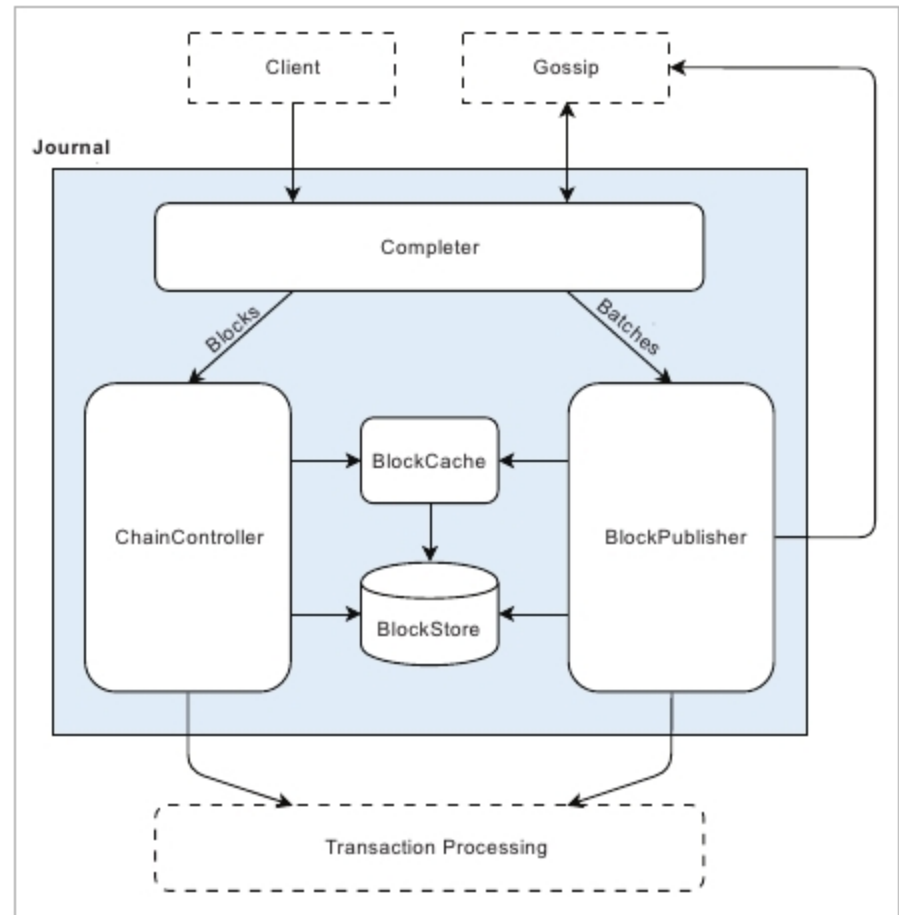


Figure 3: The journal in Sawtooth

also supports dynamic consensus. The initial consensus algorithm for the blockchain is set in the genesis block, but can be changed during a blockchain's lifetime with the settings transaction processor.

The BlockCache is an in-memory construct that is rebuilt when the system is started. It holds the working set of blocks for the validator. It tracks the processing state of each block as valid, invalid, or unknown.

The BlockStore is a persistent on-disk store of all blocks in the current chain, i.e., the list of blocks from the current chain head back to the genesis block. When the validator is started, the contents of the BlockStore are trusted to be the current 'state of the blockchain' and the blocks are considered formally complete. It also internally maps transaction-to-block and batch-to-block. These mappings can be rebuilt if they are missing or corrupt. They are stored in a format that is cached to disk, so are not held in memory at all times. Blocks stored can be accessed by the block ID. Blocks can also be accessed via batch ID, transaction ID or the block number. **END** 🐧

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